12.1 Orthogonal Polynomials

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SL ODE

$$p(x)y'' + q(x)y' + \lambda y = 0$$

$$p(x) = \alpha x^2 + \beta x + \nu \qquad q(x) = \mu x + \nu$$

$$y_n(x) = \sum_{j=0}^{n} g_j x^j$$

Subst.

$$p(x)\sum_{j=2}^{n} g_j j(j-1)x^{j-2} + q(x)\sum_{j=1}^{n} g_j j x^{j-1} + \lambda \sum_{j=0}^{\infty} g_j x^j = 0$$

$j = n$, x^n terms

$$\alpha g_n n(n-1) + \mu g_n n + \lambda g_n = 0 \Rightarrow \lambda_n = -n(n-1)\alpha - n\mu$$

Self-adjoint if $p'(x) = q(x)$

Weight function

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Otherwise $(\omega p)' = \omega q$ or $\omega' = \omega \dfrac{q - p'}{p}$

$$\omega = p^{-1} \exp\left(\int \frac{q}{p} dx \right)$$

$$\Rightarrow \frac{d}{dx}\left[\omega(x) p(x) y'\right] + \lambda \omega(x) y = 0$$

Rodrigues formula

$$y_n(x) = \frac{1}{\omega(x)} \left(\frac{d}{dx}\right)^n \left[\omega p^n(x)\right]$$

Rodrigues Formula

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Proof

$$p\left[\omega p^n\right]' = \omega' p^{n+1} + n\omega p^n = \frac{q-p'}{p}\omega p^{n+1} + n\omega p^n$$

$$= \omega p^n \left[(n-1)p' + q\right]$$

Leibniz rule

$$(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(n-k)} g^{(k)}$$

ex $\left(\frac{d}{dx}\right)^9 \left(x^2 \sin x\right) = \binom{9}{0} x^2 \left(\frac{d}{dx}\right)^9 (\sin x) + \binom{9}{1}(2x)\left(\frac{d}{dx}\right)^8 \sin x$

$$+ \binom{9}{2}(2)\left(\frac{d}{dx}\right)^7 \sin x$$

$$= x^2 \cos x + 18x \sin x - 72 \cos x$$

$$p(\omega p^n)' = \omega p^n \left[(n-1)\frac{p'}{p} + \frac{q}{p}\right]$$

Diff $n+1$ times & divide by ω

$$\frac{p}{\omega}\left(\frac{d}{dx}\right)^{n+2}[\omega p^n] + \frac{(n+1)}{\omega}p'\left(\frac{d}{dx}\right)^{n+1}[\omega p^n] + \frac{n(n+1)}{\omega}p''\left(\frac{d}{dx}\right)^{n}[\omega p^n]$$

$$= \frac{(n-1)p'+q}{\omega}\left(\frac{d}{dx}\right)^{n+1}[\omega p^n] + \frac{(n+1)\left[(n-1)p''+q'\right]}{\omega}\left(\frac{d}{dx}\right)^{n}[\omega p^n]$$

Identify $y_n = \frac{1}{\omega}\left(\frac{d}{dx}\right)^n[\omega p^n]$

$$\frac{p}{\omega}\left(\frac{d}{dx}\right)^{n+2}[\omega p^n] + \frac{2p'-q}{\omega}\left(\frac{d}{dx}\right)^{n+1}[\omega p^n] - \left[\frac{n^2-n-2}{2}p'' + (n+1)q'\right]y_n = 0$$

$\frac{d}{dx}\frac{1}{\omega}\left(\frac{d}{dx}\right)^n(\omega p^n)$

Moving $\frac{1}{\omega}$ inside

$$\omega^{-1}[f(x)]', \quad [\omega^{-1}f]' = \omega^{-1}f' + \left(\frac{d}{dx}\omega^{-1}\right)f$$

$$\omega^{-1}f' = [\omega^{-1}f]' - \left(\frac{d}{dx}\omega^{-1}\right)f$$

$$\omega^{-1}f'' = [\omega^{-1}f]'' - 2\left(\frac{d}{dx}\omega^{-1}\right)f' - \left(\frac{d^2}{dx^2}\omega^{-1}\right)f$$

$$p y_n'' + q y_n' - \frac{\left[\frac{n^2-n}{2} p'' + n q'\right] y_n = 0}{\lambda_n}$$

Ex 12.1.1 Hermite eqn $y'' - 2xy' + \lambda y = 0$

$p = 1 \quad q = -2x \Rightarrow \omega(x) = \exp\left(\int -2x\,dx\right) = e^{-x^2}$

$$y_n(x) = \frac{(-1)^n}{\omega(x)} \left(\frac{d}{dx}\right)^n [\omega p^n] = (-1)^n e^{x^2} \left(\frac{d}{dx}\right)^n e^{-x^2}$$

Schlafli integral

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Cauchy

Using Integral formula

$$y_n(x) = \frac{1}{\omega(x)} \frac{n!}{2\pi i} \oint \frac{\omega(z)\,[P(z)]^n}{(z-x)^{n+1}}\,dz$$

Generating function

$$g(x,t) = \sum_n c_n f_n(x)\, t^n$$

$n \geq 0$ Taylor series

$-\infty < n < \infty$ Laurent series

Applying Residue Thm $\quad c_n f_n(x) = \frac{1}{2\pi i} \oint \frac{g(x,z)}{z^{n+1}}\,dz$

Ex 12.1.2 Hermite Polynomials

$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$$

$$\frac{\partial}{\partial t} e^{-t^2+2tx} = 2(x-t)e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{H_n(x)}{(n-1)!} t^{n-1}$$

$$2(x-t)\sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n = \cdots$$

$$\sum_{n=0}^{\infty} 2x \frac{H_n(x)}{n!} t^n - \sum_{n=0}^{\infty} 2\frac{H_n(x)}{n!} t^{n+1} = \sum_{n=0}^{\infty} \frac{H_n(x)}{(n-1)!} t^{n-1}$$

Recurrence

$$2x\frac{H_n(x)}{n!} - 2\frac{H_{n-1}(x)}{(n-1)!} = \frac{H_{n+1}(x)}{n!} \Rightarrow 2xH_n(x) - 2nH_{n-1}(x) = H_{n+1}(x)$$

$$\frac{\partial}{\partial x} e^{-t^2+2xt} = 2t \; e^{-t^2+2xt} = \sum_{n=0}^{\infty} 2 \frac{H_n(x)}{n!} t^{n+1} = \sum_{n=0} \frac{H_n'(x)}{n!} t^n$$

$$H_n'(x) = 2n \, H_{n-1}(x)$$

Finding Generating Function

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$$g(x,t) = \frac{1}{w(x)} \sum_{n=0}^{\infty} c_n t^n \quad \frac{n!}{2\pi i} \oint \frac{w(z)[p(z)]^n}{(z-x)^{n+1}} dz$$

$c_n = \frac{(-1)^n}{n!}$

Ex Hermite eqn $w(x) = e^{-x^2}$ $\quad p(x) = 1$

$$g(x,t) = e^{x^2} \sum_{n=0}^{\infty} \frac{t^n}{2\pi i} c_n n! \oint \frac{e^{-z^2}}{(z-x)^{n+1}} dz$$

$$= e^{x^2} \frac{1}{2\pi i} \oint dz \; \frac{e^{-z^2}}{z-x} \sum_{n=0}^{\infty} \frac{t^n}{(z-x)^n} c_n n!$$

$$= e^{x^2} \frac{1}{2\pi i} \oint dz \; \frac{e^{-z^2}}{z-x} \frac{1}{1+\frac{t}{z-x}} = e^{x^2} \frac{1}{2\pi i} \oint \frac{e^{-z^2}}{z-(x-t)} dz$$

$$= e^{x^2} e^{-(x-t)^2} = e^{-t^2+2tx}$$

Ex 12.1.3

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Legendre Eqn $(1-x^2)y'' - 2xy' + \lambda y = 0$ $p(x) = 1-x^2$ $\omega(x)$

$$g(x,t) = \sum_{n=0}^{\infty} \frac{t^n}{2\pi i} c_n n! \oint \frac{(1-z^2)^n}{(z-x)^{n+1}} dz = \frac{1}{2\pi i} \oint \frac{dz}{z-x} \sum_{n=0}^{\infty} t^n \frac{(1-z^2)^n}{(z-x)^n} c_n n!$$

$$= \frac{1}{2\pi i} \oint \frac{dz}{z-x} \sum_{n=0}^{\infty} (-1)^n \left[\frac{(1-z^2)t}{2(z-x)}\right]^n \qquad c_n = \frac{(-1)^n}{2^n n!}$$

$$= \frac{-2}{t} \frac{1}{2\pi i} \oint \frac{1}{z^2 - \frac{2z}{t} + \frac{2x-t}{t}} dz = \frac{-2}{t} \frac{1}{2\pi i} \oint \frac{1}{(z-z_1)(z-z_2)} dz$$

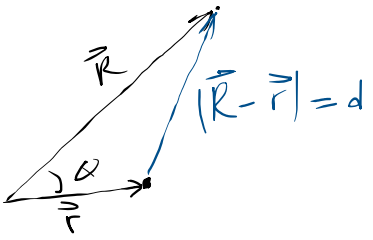
$$z_{1,2} = \frac{1}{t} \mp \frac{\sqrt{1-2xt+t^2}}{t}$$

$$= \frac{-2}{t} \frac{1}{z_1 - z_2} = \frac{1}{\sqrt{1-2xt+t^2}}$$

Ex Potential expansion

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$$V = \frac{K}{d} = \frac{K}{|\vec{R}-\vec{r}|} = \frac{K}{\sqrt{R^2 - 2rR\cos\theta + r^2}}$$



$$= \frac{K}{R} \frac{1}{\sqrt{1 - 2\left(\frac{r}{R}\right)\cos\theta + \left(\frac{r}{R}\right)^2}}$$

$$= \frac{K}{R} \frac{1}{\sqrt{1 - 2xt + t^2}} = \frac{K}{R} \sum_{n=0}^{\infty} P_n(x)\, t^n$$

$$= \frac{K}{R} \sum_{n=0}^{\infty} P_n(\cos\theta)\left(\frac{r}{R}\right)^n = K \sum_{n=0}^{\infty} \frac{r^n}{R^{n+1}} P_n(\cos\theta)$$

$(\vec{R}-\vec{r})\cdot(\vec{R}-\vec{r})$

$t = \frac{r}{R}$

$x = \cos\theta$

Ch 15 Legendre functions

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$$-\nabla^2\psi + V(r)\psi = \lambda\psi$$

$$(1-x^2)P''(x) - 2xP'(x) - \frac{m^2}{1-x^2}P(x) + l(l+1)P(x) = 0$$

Associated Legendre Eq.

$m=0 \Rightarrow$ Legendre ODE

15.1 Legendre Poly

$$(1-x^2)P''(x) - 2xP'(x) + \lambda P(x) = 0$$

Reg. Singular points at $x = \pm 1, \infty$

$\Rightarrow$ Series solution about $x=0 \Rightarrow$ radius of convergence of 1

$$P_l(x) \iff \lambda_l = l(l+1)$$

$$g(x,t) = \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)\, t^n$$

$$g(1,t) = \frac{1}{1-t} = \sum_{n=0}^{\infty} t^n = \sum_{n=0}^{\infty} P_n(1)\, t^n \quad \Rightarrow P_n(1) = 1$$

$$g(x,t) = g(-x,-t) = \sum_{n=0}^{\infty} P_n(-x)(-1)^n t^n \quad \Rightarrow P_n(-x) = (-1)^n P_n(x)$$

$$P_n(-1) = (-1)^n$$

$$g(0,t) = (1+t^2)^{-1/2} = \sum_{n=0}^{\infty} \binom{-1/2}{n} t^{2n} = \sum_{n=0}^{\infty} P_n(0)\, t^n$$

$$(1+x)^m = \sum_{k=0}^{m} \binom{m}{k} x^k$$

$$\binom{m}{k} = \frac{m(m-1)\cdots(m-k+1)}{k!}$$

$$P_{2n}(0) = \binom{-1/2}{n} = \frac{(-1/2)(-\frac{3}{2})(-\frac{5}{2})\cdots(-\frac{1}{2}-n+1)}{n!}$$

$$= \frac{(-1)^n}{2^n} \frac{(1)(3)(5)\cdots(2n-1)}{n!} = (-1)^n \frac{(2n-1)!!}{(2n)!!}$$

$$(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} \binom{-1/2}{n} (-2xt+t^2)^n = \sum_{n=0}^{\infty} P_n(x)\, t^n$$

Coefficient of x^n $\quad (-2)^n \binom{-1/2}{n} = \dfrac{(2n-1)!!}{n!}$

Property:

$P_n(x)$ has scaling and sign $\quad P_n(1) = 1 \qquad P_n(-1) = (-1)^n$

$P_{2n}(x)$ even $\qquad P_{2n+1}$ odd

$P_{2n}(0) = (-1)^n \dfrac{(2n-1)!!}{(2n)!!} \qquad P_{2n+1}(0) = 0$

$P_n(x)$ has degree n, coeff of x^n is $\dfrac{(2n-1)!!}{n!}$

$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{2^n k!\,(n-k)!\,(n-2k)!}\, x^{n-2k}$$

HW

Recurrence Rel

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$$\frac{\partial g}{\partial t} = \frac{x-t}{(1-2xt+t^2)^{3/2}} = \sum_{n=0}^{\infty} n P_n(x) t^{n-1}$$

$$\frac{x-t}{(1-2xt+t^2)^{1/2}} = (1-2xt+t^2) \sum_{n=0}^{\infty} n P_n(x) t^{n-1}$$

$$(x-t) \sum_{n=0}^{\infty} P_n(x) t^n = (1-2xt+t^2) \sum_{n=0}^{\infty} n P_n(x) t^{n-1}$$

$$(2n+1) x P_n(x) = (n+1) P_{n+1}(x) + n P_{n-1}(x) \quad , n = 1, 2, 3, \ldots$$

$$\frac{\partial g}{\partial x} = \frac{t}{(1-2xt+t^2)^{3/2}} = \sum_{n=0}^{\infty} P_n'(x) t^n \Rightarrow P_{n+1}'(x) + P_{n-1}'(x) = 2x P_n'(x) + P_n(x)$$

Rodrigus Formula

$$P_n(x) = \frac{1}{2^n n!} \left(\frac{d}{dx}\right)^n (x^2-1)^n$$

coefficient of $x^n \Rightarrow$ $\frac{1}{2^n n!}$ $\frac{\partial^n}{\partial x^n} x^{2n}$ $= \frac{1}{2^n n!}(2n)(2n-1)\cdots(n+1)$

$$= \frac{1}{2^n n!} \frac{2n!}{n!} = \frac{(2n-1)!!}{n!}$$

15.2 Orthogonality

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$$\int_{-1}^{1} P_n(x) P_m(x)\, dx = 0 \qquad (n \neq m)$$

Normalization

$$\frac{1}{1-2xt+t^2} = \left[\sum_{n=0}^{\infty} P_n(x)\, t^n\right]^2 \Rightarrow \int_{-1}^{1} \frac{dx}{1-2xt+t^2} = \sum_{n=0}^{\infty} t^{2n} \int_{-1}^{1} [P_n(x)]^2\, dx$$

$u = 1-2xt+t^2 \Rightarrow du = -2t\,dx \Rightarrow \frac{1}{2t}\int_{(1-t)^2}^{(1+t)^2} \frac{du}{u} = \frac{1}{t}\ln\left(\frac{1+t}{1-t}\right) = \sum_{n=0}^{\infty} \frac{2}{2n+1} t^{2n}$

HW

$$\int_{-1}^{1} P_n(x) P_m(x)\, dx = \frac{2}{2n+1}\delta_{mn}$$

Legendre Series

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$$f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$$

$$a_n = \frac{2n+1}{2} \int_{-1}^{1} P_n(x) f(x) \, dx$$

Associated Legendre Eqn.

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$$(1-x^2)P''(x) - 2xP'(x) + \left[\lambda - \frac{m^2}{1-x^2}\right]P(x) = 0$$

$$P = (1-x^2)^{m/2}\mathcal{P}$$

$$(1-x^2)\mathcal{P}'' - 2x(m+1)\mathcal{P}' + [\lambda - m(m+1)]\mathcal{P} = 0$$

Frob. Method $\Rightarrow \lambda = \ell(\ell+1) \Rightarrow \mathcal{P}_\ell^m \Rightarrow P_\ell^m = (1-x^2)^{m/2}\mathcal{P}_\ell^m(x)$

$$\mathcal{P}_\ell^m = (-1)^m \frac{d^m}{dx^m} P_\ell(x) \qquad P_\ell^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x)$$

$$P_\ell^0 = P_\ell \qquad 0 \leq m < \ell$$

Rodrigues Formula

$$P_\ell^m(x) = \frac{(-1)^m}{2^\ell \ell!}(1-x^2)^{m/2} \frac{d^{\ell+m}}{dx^{\ell+m}}(x^2-1)^\ell$$

Negative m

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x)$$

HW

Generating Function of $P_l^m(x)$

$$g_m(x,t) = \frac{(-1)^m (2m-1)!!}{(1-2xt+t^2)^{m+\frac{1}{2}}} = \sum_{n=0}^{\infty} P_l^m(x)\, t^n$$

Orthogonality

$$\int_{-1}^{1} P_p^m(x) P_q^m(x)\, dx = \frac{2}{2p+1} \frac{(p+m)!}{(p-m)!} \delta_{pq}$$

$$\int_{-1}^{1} P_l^m(x) P_l^n(x) (1-x^2)^{-1} dx = \frac{(l+m)!}{m(l-m)!} \delta_{mn}$$

Spherical Harmonics

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$\Theta(\theta)\,\Phi(\varphi)$

$\Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}$ $\quad m = \ldots -2, -1, 0, 1, 2, \ldots$ $\quad \Rightarrow \int_0^{2\pi} \Phi_m^* \Phi_n \, d\varphi = \delta_{mm'}$

$\Theta_{lm}(\cos\theta) = \sqrt{\frac{(2l+1)}{2}\frac{(l-m)!}{(l+m)!}}\, P_l^m(\cos\theta) \Rightarrow \int_0^{\pi} \Theta_{lm}^* \Theta_{l'm} \sin\theta \, d\theta = \delta_{ll'}$

$Y_l^m(\theta,\varphi) = \sqrt{\frac{(2l+1)}{2}\frac{(l-m)!}{(l+m)!}}\, P_l^m(\cos\theta)\, \frac{e^{im\varphi}}{\sqrt{2\pi}}$

$\int_0^{2\pi} d\varphi \int_0^{\pi} \sin\theta \, d\theta \; Y_l^{m*} Y_l^{m'} = \delta_{ll'}\,\delta_{mm'}$